Transformers in the World

The family, the LM head, and what it costs

Before we start: videos and a demo

Recommended viewing

Hugging Face's course, three short videos (16 min in all): Transformer models: Encoders, Transformer models: Decoders, Transformer models: Encoder-Decoders. Today's first section.

3Blue1Brown, Transformers, the tech behind LLMs | Deep Learning Chapter 5, from 20:22: the unembedding, then softmax with temperature (22:22). Today's second section, and LLM-6's opening.

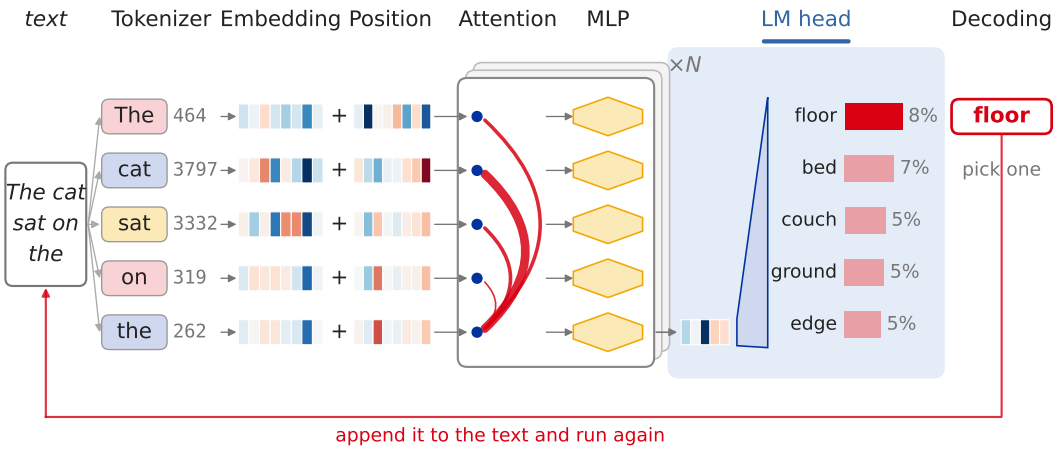
Live demo: bbycroft.net/llm

Brendan Bycroft's LLM Visualization: every matrix of a transformer in 3D. A guided walk through a tiny model, then GPT-2 small, GPT-2 XL and GPT-3, drawn to scale.

Try:

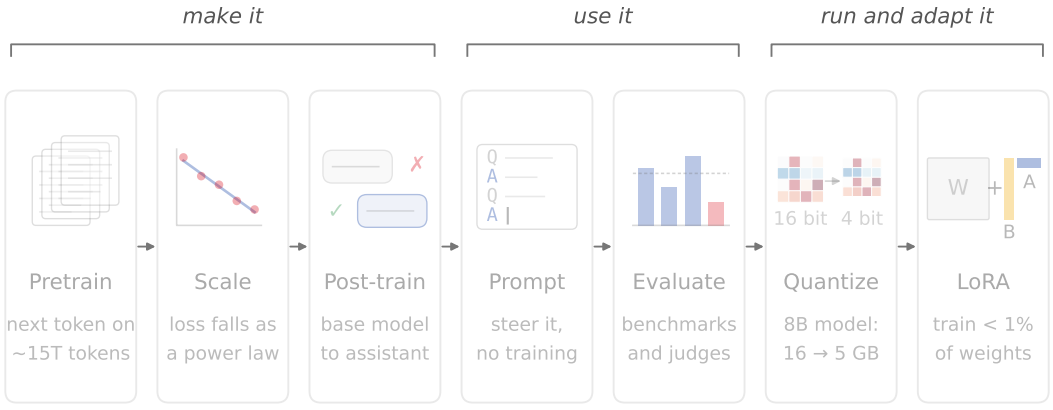
- the walk-through, up to "Output"
- GPT-2 small next to GPT-3: the same block, bigger
- find the LM head at the very end

Where we are: one forward pass



Today: the **LM head** (last vector to a distribution) and three ways to wire the block.

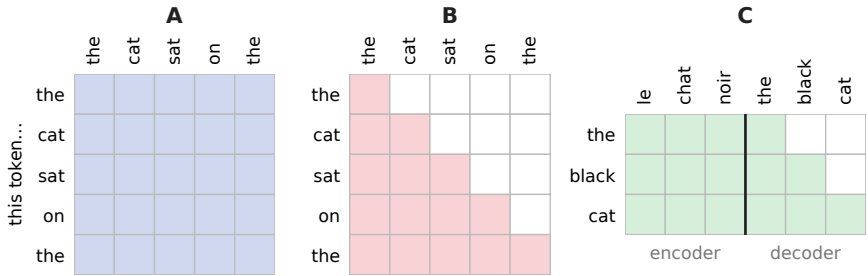
Where we are: the life of a model



Later: where these weights come from, and what you do with them once you have them.

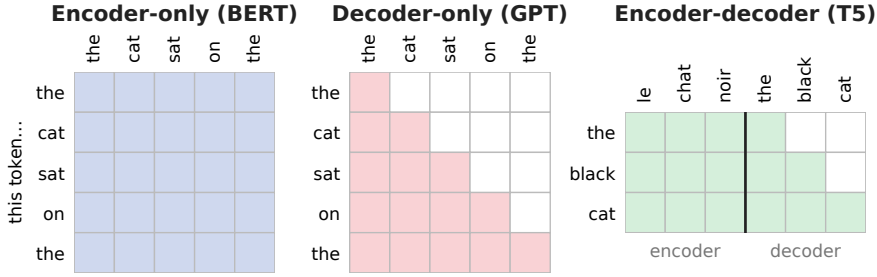
Which of these can write text?

Three attention masks for the same block. A row is the token being computed; the shaded cells are the tokens it may look at.



Predict first: which of A, B, C could *write* a sentence, left to right, one token at a time? More than one answer is allowed.

B and C can. A cannot.



A lets the first "the" look at "sat". When you write, word 3 does not exist yet while you choose word 2, so a model trained to peek cannot be run forwards. **B** is LLM-3's causal mask. **C** writes English with a causal mask too, and also reads the whole French sentence (drawn as one grid; in the model it is two attention layers, one over each sequence).

The block is the same in all three. The difference is one decision: **who is allowed to look at whom?**

Outline

The family

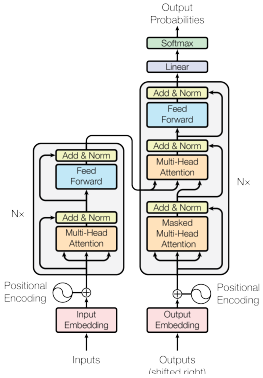
The LM head

Why it won, and what it costs

The family

One block, three ways of wiring it.

The original: an encoder and a decoder



Vaswani et al. (2017), Figure 1

The 2017 paper was about **translation**, so it has two stacks:

Encoder – reads the source sentence. Every token may see every other token.

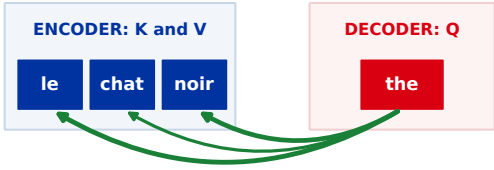
Decoder – writes the target sentence, one token at a time, so it is **causally masked** (LLM-3).

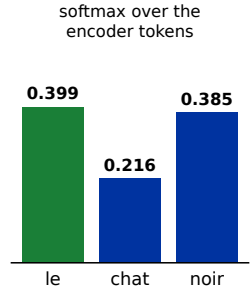
They are joined by **cross-attention**: the decoder supplies the queries, and the *encoder's* output supplies the keys and values.

Self-attention asks "what in *this* sentence matters to me?" Cross-attention asks "what in *that other* sequence matters to me?" Identical mechanism; the keys and values just come from somewhere else.

Cross-attention, with actual numbers

The decoder is writing an English sentence and has produced "the". The encoder has already read the French source. Where should "the" look?





One head, $d_k = 4$, toy random weights. Each score is $q \cdot k/\sqrt{4}$: for "le", $0.6788/2 = 0.3394 \approx +0.339$; the others are -0.273 and $+0.304$. Softmax gives **0.399**, **0.216**, **0.385** (sum 1), and the output is $0.399\, v_{\text{le}} + 0.216\, v_{\text{chat}} + 0.385\, v_{\text{noir}}$, exactly as in LLM-3.

What actually changed

Compare with the self-attention you computed in LLM-3. The formula is character-for-character the same:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V$$

	Self-attention	**Cross-attention**
Q comes from	this sequence	**the decoder**
K, V come from	this sequence	**the encoder**

Where it sits: every decoder block has **three** sublayers - masked self-attention, then cross-attention (its own W_Q, W_K, W_V; the keys and values come from the encoder's *final* output, the same for every decoder block), then the MLP.

That is the entire difference: **two of the three inputs come from somewhere else**. No new mechanism, no new maths – which is why the original paper could reuse the same block for both.

Three architectures



same block, three ways of deciding who may look at whom

Nobody changed the block. **Encoder-only** sees in both directions and is good at understanding: BERT is trained to fill in hidden words (masked language modeling, LLM-3). **Decoder-only** sees only backwards, which makes it able to generate. **Encoder-decoder** keeps both and joins them with cross-attention.

The family, by name

- **Encoder-only**: BERT (Google, 2018). Not retired: ModernBERT (December 2024) was trained on 2 trillion tokens, reads 8,192 tokens at once, and is built for classification and retrieval - the embeddings that retrieval-augmented generation (RAG) needs.
- **Encoder-decoder**: T5 (Raffel et al., 2020) turns every task into text in, text out, with a prefix that names the task: "translate English to German: That is good." $\rightarrow$ "Das ist gut."; "mnli premise: I hate pigeons. hypothesis: My feelings towards pigeons are filled with animosity." $\rightarrow$ "entailment". Pretrained by blanking out spans of text and writing them back.
- **Decoder-only**: GPT-2, and every chat model you have used.

A running joke of 2018-2020: ELMo (2018), BERT (2018), ERNIE (2019), Grover (2019), KERMIT (2019), Big Bird (2020). For a while, language models were named after Sesame Street.

Why decoder-only won the last five years

- **One objective, every token.** BERT's fill-in-the-blank needs no labels either, but it learns only from the 15% of tokens it hides, and it cannot write. Next-token prediction learns from every token (LLM-3's mask), and the model it trains writes.
- **It subsumes the others.** "Translate this: ..." is just a prompt. Tasks that needed a separate architecture became a formatting question.
- **It is the simplest thing to scale.** One stack, one loss, no encoder to keep in sync.

This is a statement about the last few years, not a theorem. Encoder-only models are still the right tool when you want an *embedding* rather than a continuation – which is exactly what the retrieval chapter (RAG, after LLM-11) will use them for.

Mask A writes after all: diffusion language models

- Mask A failed because word 3 does not exist yet while word 2 is chosen. A diffusion language model makes every position exist from the start: the answer has a fixed length and begins as all blanks. That is BERT's fill-in-the-blank, run again and again.
- Each step predicts every blank at once, keeps the guesses the model is most sure of (highest probability), blanks the rest again, and repeats for a set number of steps.
- LLaDA (Nie et al., NeurIPS 2025): 8B parameters, "does not use a causal mask"; "competitive with" LLaMA 3 8B. The selling point is speed, many tokens per step: Mercury (2025) and Google's open DiffusionGemma (2026), with quality a step behind.

The cold open's answer holds for writing one token at a time. Write everything at once and refine it, and mask A works - so far trading some quality for speed.

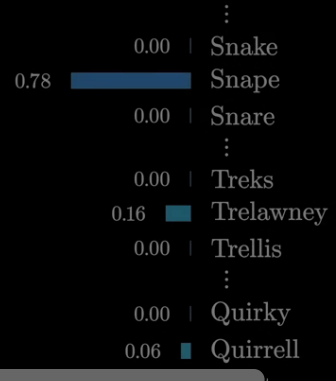
Nie et al., arXiv 2502.09992; Inception Labs (Feb 2025: over 1,000 tokens/s on one GPU); Google (10 June 2026: up to 4× faster, quality below Gemma 4).

The LM head

From the last vector to a probability for every token.

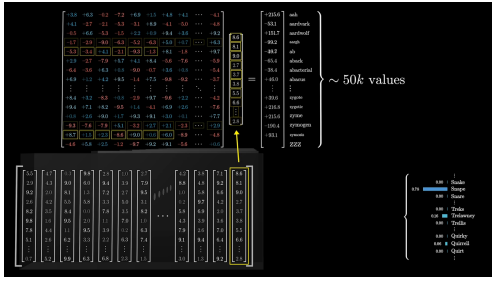
Harry Potter was a highly unusual boy ... least favourite teacher, Professor ???





Next-token prediction, in practice: the model outputs a distribution over the whole vocabulary, and one continuation is far more likely than the rest.

From a vector to a word



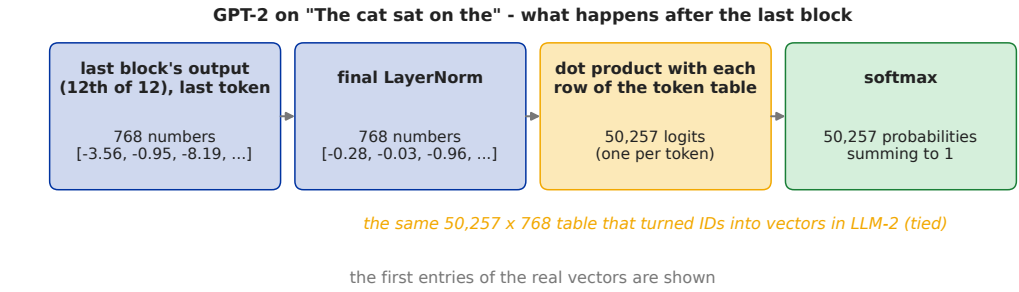
3Blue1Brown

Every position leaves the last block with a vector. In training, every one of them predicts its own next token at once (LLM-3's mask keeps that honest). When generating, the earlier next tokens are already written, so only the **last** vector's prediction is new. After one final LayerNorm it is multiplied by an **unembedding matrix** – one row per vocabulary entry, so roughly 50,000 numbers come out.

Those are **logits**. A softmax turns them into a probability distribution, and you sample from it. That is the entire output stage.

The model produces a *distribution* over every word, not a word. Choosing is a separate decision: temperature and the other decoding strategies (LLM-6).

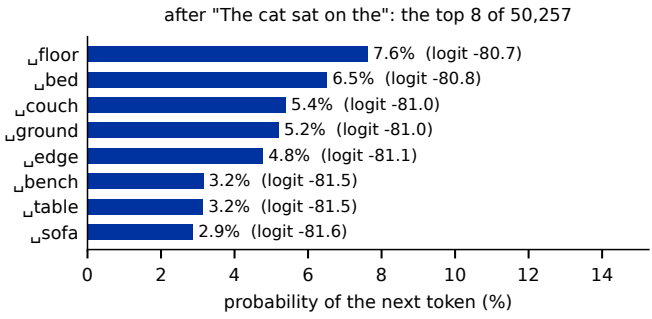
GPT-2's head, on the map's prompt



The "unembedding matrix" in GPT-2 is not a new matrix. It is the token table from LLM-2, used a second time: **tied**. Row j turned token j into a vector on the way in; on the way out, its dot product with the last vector scores token j. Both jobs want tokens used alike to get vectors alike (LLM-2).

The table is $50{,}257 \times 768 = 38.6\text{M}$ numbers, **31%** of GPT-2's 124.4M parameters. Tying it means the head costs no parameters of its own.

What comes out



Real output of the cached GPT-2 small. (␣ is the leading space, part of the token: LLM-1.)

The top five hold **30%** of the probability. Only **14** of the 50,257 tokens get more than 1%; the remaining mass is spread thinly over everything else.

And look at the logits: **all of them are near** -80.

Predict: if every logit is around -80, how can "floor" end up with 7.6%?

One logit, by hand

A logit is one dot product, 768 terms long: the last vector against one row of the token table. For the token "floor":

$$z(\text{floor}) = (-0.284)(-0.122) + (-0.033)(0.161) + (-0.962)(-0.025) + (0.225)(-0.024) + \dots$$

The first four terms give 0.048; all 768 give $z(\text{floor}) = -80.65$. Likewise $z(\text{bed}) = -80.81$.

Every row of the table gets its own dot product: 50,257 of them, which is one matrix multiplication. A row that points more the same way as the last vector gets a **higher** score than the others - here they all sit around -80, and only the differences count.

Every logit is near -80. Back to the question: how does "floor" end up with 7.6%?

Answer: only differences matter

Two tokens: compare them directly.

$$\frac{p(\text{floor})}{p(\text{bed})} = \frac{\exp z(\text{floor})}{\exp z(\text{bed})} = \exp\big(z(\text{floor}) - z(\text{bed})\big) = \exp(0.157) = 1.17$$

Check against the bars: $7.64\% \,/\, 6.53\% = 1.17$.

One token: compare it with all of them.

$$p(\text{floor}) = \frac{\exp z(\text{floor})}{\sum_j \exp z_j} = \exp\big(z(\text{floor}) - \log \textstyle\sum_j \exp z_j\big) \qquad (j \text{ runs over all } 50{,}257 \text{ tokens})$$

$$= \exp\big(-80.654 - (-78.082)\big) = \exp(-2.572) = \mathbf{7.64}\%$$

$\log \sum_j \exp z_j = -78.08$ sits *above* the largest logit (-80.65): 50,257 positive terms add up to more than the largest one alone.

Add the same number c to every logit and nothing changes: $e^{z_i + c} / \sum_j e^{z_j + c} = e^{z_i} / \sum_j e^{z_j}$, because the factor e^c cancels. The softmax only sees **differences**. That is why -80 is harmless, and it is the knob LLM-6 turns: temperature stretches or shrinks those differences.

By hand: three logits near −80

Three tokens with logits $(-80,\ -81,\ -82)$. Subtract the largest from all three: $(0,\ -1,\ -2)$.

$$\text{softmax} = \frac{(e^0,\ e^{-1},\ e^{-2})}{e^0 + e^{-1} + e^{-2}} = \frac{(1,\ 0.37,\ 0.14)}{1.50} = (\mathbf{0.67},\ 0.24,\ 0.09)$$

The same as for $(-80, -81, -82)$, because the shift cancels. That is also how code computes a softmax safely: subtract the largest logit first (LLM-3's ten lines did), so no e^z overflows or underflows.

Back to GPT-2. The denominator's log, $\log \sum_j e^{z_j} = -78.08$, has a name: **log-sum-exp**. It sits 2.57 above floor's -80.65, so all 50,257 tokens together carry $e^{2.57} = 13.1$ times floor's weight, and floor gets $1/13.1 = 7.6\%$.

Floor is the single favourite, but the other 50,256 tokens together outweigh it about 12 to 1. LLM-6 started from exactly that tail.

Why it won, and what it costs

Every architecture buys something with something.

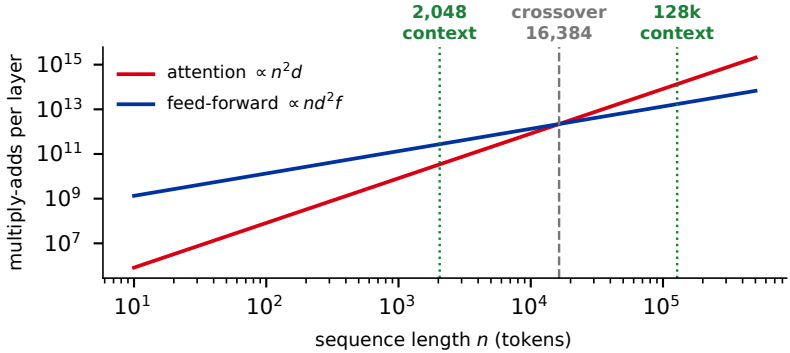
Why transformers won

1. **It parallelises.** An RNN had to walk the sequence in order. Attention is a few big matrix multiplications over the whole sequence at once (LLM-3), which is exactly the workload a GPU wants. That is training; generating is still one token at a time (LLM-6).
2. **Because it parallelises, it scales.** You can afford more data and more parameters.
3. **Because it scales, it kept getting better.** The clearest example is **in-context learning**: give a model a few examples in the prompt and it follows the pattern, *with no weight updates at all* ("cat → chat, dog → chien, bird →" gets "oiseau"). Small models do it badly, large ones well; whether that improvement is sudden or gradual is debated (LLM-8).

Notice the shape of that argument: the win is not that attention is a cleverer model of language. It is that attention is a **better fit for the hardware**, and everything else followed from being able to train bigger. Architecture that suits the machine beats architecture that suits your intuition.

The bill: attention is quadratic

Every token attends to every token, so the score matrix is $n \times n$. Double the context and you quadruple that cost.



$d_{\text{model}} = 4096$, $f = 4$. The grid costs memory too: at 128,000 tokens one head's $n \times n$ scores are 16.4 billion numbers, so FlashAttention never stores it whole (LLM-14).

Read that graph carefully

The usual story is "attention is the bottleneck". The arithmetic says: **not until the sequence gets long**.

You can work out where it flips in two lines. Attention's $n \times n$ part costs about $n^2 d$ (only the $n \times n$ part; the Q, K, V and output projections grow like the feed-forward and would only push the crossover further out); the feed-forward costs about $nd^2 f$, where f is the widening factor (usually 4). Set them equal:

$$n^2 d = nd^2 f \quad \Longrightarrow \quad \boxed{n = d \cdot f}$$

So for $d_{\text{model}} = 4096$ and $f = 4$, the crossover is at $4096 \times 4 = \mathbf{16{,}384}$ tokens. Halve the model and you halve the crossover. For GPT-2 small, $768 \times 4 = 3{,}072$: three times its whole 1,024-token window.

So for a 2,048-token context, optimising attention is optimising the smaller half. "Quadratic" describes how the cost *grows*, not where the cost *is* at every size. FlashAttention and sparse variants attack the n^2 term; 2025 hybrids keep full attention in a quarter of their layers or fewer (LLM-14).

The honest limit: almost no inductive bias

An **inductive bias**: what a model assumes before it sees any data. Every model here had one:

Model	Built-in assumption	Consequence
Decision tree	axis-aligned splits matter	works on tiny tabular data
CNN	nearby pixels are related	learns vision from thousands of images
RNN	order matters, recent matters more	handles sequences by construction
Transformer	**almost nothing**	**needs enormous data**

A Transformer does not know that neighbouring words are related, or even that words have an order – we had to *bolt position on* in LLM-2. Its only built-in assumptions are the ones we add by hand: positions, and the causal mask.

That weakness is the same fact as its strength. Assuming nothing means it can learn any structure, and **must** learn all of it from data. This is why a ViT loses to a CNN on a small image dataset and wins on a huge one (Dosovitskiy et al., 2021: behind ResNets on ImageNet alone, ahead after 14-300M images), and why transformers arrived only once the data and the compute did.

Same block, everywhere

Domain	What became a token	Chapter
Text	subword pieces	this chapter
Images (ViT)	16×16 patches	vision-language models
Audio	spectrogram frames or codec codes	audio
Proteins	amino acids	protein language models, AlphaFold
Actions	robot commands	vision-language-action

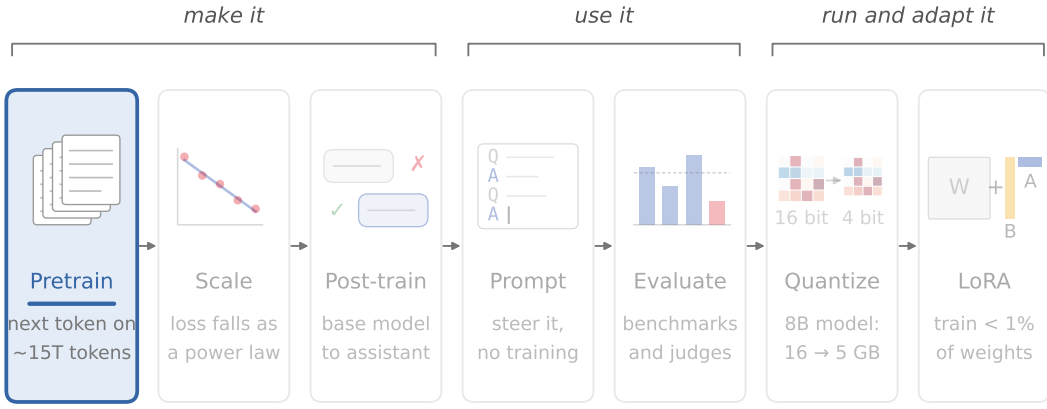
Nobody redesigned the block for any of these. They redesigned what counts as a **token**. That is the practical legacy of the architecture: if you can serialise your problem into a sequence of vectors, you can throw a Transformer at it.

AlphaFold 2 is built on attention. Half of the 2024 Nobel Prize in Chemistry went to its makers, Demis Hassabis and John Jumper, "for protein structure prediction".

Recap

- **One block, three wirings.** Encoder-only (BERT) understands, decoder-only (GPT) generates, encoder-decoder (T5) translates. The difference is who may look at whom - the mask, plus cross-attention in the decoder - not the attention itself.
- **Cross-attention** is ordinary attention with the keys and values taken from another sequence. Diffusion LMs write with no causal mask at all, many tokens per step.
- **The LM head** is one dot product per vocabulary entry with the token table (tied in GPT-2), then a softmax. The output is a distribution, not a word, and only the differences between logits matter.
- **It won because it fits the GPU**, not because it is a better theory of language.
- **It is quadratic in sequence length**, but at $d_{\text{model}} = 4096$ that only dominates past 16,384 tokens.
- **It assumes almost nothing**, so it can learn anything and needs vast data to do it.

Next on the map



Next (LLM-7): where these probabilities come from - pretraining.

Sources & further reading

- **Watch:** Hugging Face course (2021), "Transformer models: Encoders / Decoders / Encoder-Decoders"; 3Blue1Brown (2024), Deep Learning Chapter 5. **Play:** bbycroft.net/llm.
- Vaswani et al. (2017), "Attention Is All You Need", NeurIPS (the encoder-decoder, cross-attention).
- Devlin et al. (2019), "BERT", NAACL; Raffel et al. (2020), "Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer" (T5), JMLR; Warner et al. (2025), "ModernBERT", ACL.
- Radford et al. (2019), "Language Models are Unsupervised Multitask Learners" (GPT-2).
- Nie et al. (2025), "Large Language Diffusion Models" (LLaDA), NeurIPS.
- Dosovitskiy et al. (2021), "An Image is Worth 16x16 Words" (ViT), ICLR; Jumper et al. (2021), "Highly accurate protein structure prediction with AlphaFold", *Nature*.
- Qwen team (2025), Qwen3-Next model card; Moonshot AI (2025), "Kimi Linear"; NVIDIA (2025), "Nemotron-H".

GPT-2's head: `py_src/head_figs.py` (cached GPT-2 small, raw results in `results/head_gpt2.json`).